TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Resistance of a Galvanometer by the Half-Deflection Method

Shunting the galvanometer until its deflection exactly halves — and reading its own resistance out of the shunt, plus its figure of merit.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 1 · LEVEL 1 TERM 2

1. Aim and theory

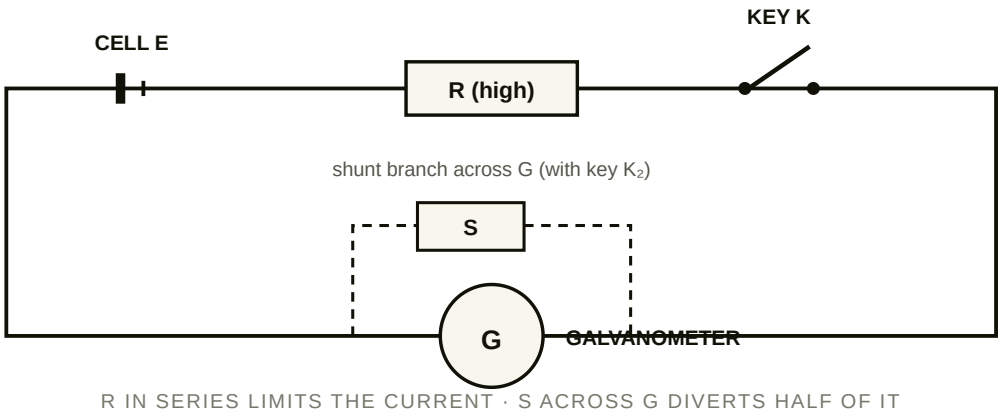
Aim: to find the resistance G of a moving-coil galvanometer, and its figure of merit, by the half-deflection method. The galvanometer is placed in series with a large resistance R and a cell of emf E. The current through it, $I_g = E/(R + G)$, produces a deflection θ. A low resistance S (the **shunt**) is then connected across the galvanometer and adjusted until the deflection falls to exactly θ/2 — that is, until the galvanometer carries exactly half its former current. Working the two circuit states through (the shunt diverts current, but R is so large that the total current is nearly unchanged) gives the elegant result

$$G = RS / (R - S) \approx S \quad \text{when } R \gg S$$

The **figure of merit** k is the current per unit deflection — the current that produces one scale division:

$$k = E / ((R + G)\,\theta) \quad \text{ampere per division}$$

Its reciprocal is the galvanometer's current sensitivity. The half-deflection trick is used because a galvanometer cannot be put in an ordinary resistance-measuring circuit: its resistance is small, it is delicate, and it must be measured while it is working as a galvanometer.



R IN SERIES LIMITS THE CURRENT · S ACROSS G DIVERTS HALF OF IT

Figure 1 — The half-deflection circuit. With K_2 open the galvanometer reads θ; closing K_2 and trimming S until the reading is θ/2 makes the shunt carry the other half.

2. Apparatus and procedure

- **Apparatus:** moving-coil galvanometer, high resistance box (up to ~10 kΩ), low resistance box (fractions of an ohm to ~100 Ω) for the shunt, cell or battery eliminator, two keys, connecting wires.
- Connect cell, key K_1, resistance box R and galvanometer in series (Fig. 1), with the shunt branch — key K_2 and resistance box S — wired across the galvanometer only.
- With K_2 open, choose R so the deflection is large and a whole number of divisions (say θ = 30). Record R and θ.
- Close K_2. Decrease S step by step until the deflection is exactly θ/2. Record S. (If the half point is jumped over, raise R slightly and repeat.)
- Repeat for several values of R (θ = 28, 26, 24…) and average the computed G.
- Compute the figure of merit from k = E/((R + G)θ) for each run and average.

3. Worked calculation

One run: E = 2 V, R = 2,000 Ω gives θ = 30 divisions; the shunt that halves it is S = 10 Ω.

$$G = RS/(R - S) = (2000 \times 10)/(2000 - 10) = 20{,}000/1990 \approx 10.05\ \Omega$$

$$k = E/((R + G)\theta) = 2/((2010)(30)) \approx 3.3 \times 10^{-5}\ \text{A/division}$$

Note how G lands just above S — the correction term matters only because R, though large, is finite. Results are tabulated run by run (R, θ, S, G, k) and the means reported.

4. Precautions

- Start with R at maximum and reduce it gradually — never connect the galvanometer to the cell without the protective series resistance.
- The emf must stay constant throughout a run: use a steady cell, and keep K_1 closed only while reading (a drifting E ruins the exact halving).
- Choose θ even and comfortably large so that θ/2 is read exactly.
- Tap the table gently / wait for the needle to settle; read the scale without parallax (mirror scale: align needle with its image).

Practice (viva preparation)

1. Why must R be large compared with G and S for this method to be clean?
2. In a run, R = 5,000 Ω and the halving shunt is S = 12 Ω. Find G.
3. Why is the deflection made to fall to exactly half — what goes wrong with "about half"?
4. A galvanometer of G = 10 Ω and figure of merit 3.3×10^{-5} A/div is to be used as a voltmeter of range 3 V. What series resistance is needed?
5. Why can the galvanometer's resistance not be found with a metre bridge in the ordinary way?

Answer key

1. Connecting the shunt changes the total circuit resistance (the G‖S combination replaces G). If R dominates the circuit, the total current is almost the same with and without the shunt, so halving the galvanometer current means the shunt truly carries an equal share — the condition the formula is built on. The exact formula $G = RS/(R - S)$ even corrects for the finite R.
2. $G = (5000 \times 12)/(5000 - 12) = 60{,}000/4988 \approx 12.03$ Ω.
3. The formula assumes I_g falls by the exact factor 2, which fixes the current division between G and S exactly. "About half" leaves the ratio unknown, and every percent of error in the ratio passes straight into G.
4. Full-scale current = k × (number of divisions). If the scale has, say, 30 divisions: $I_{fs} = 3.3 \times 10^{-5} \times 30 \approx 1.0 \times 10^{-3}$ A. For 3 V full scale, total resistance $= 3/10^{-3} = 3{,}000$ Ω, so the series (multiplier) resistance = 3,000 − 10 = 2,990 Ω.
5. A metre bridge drives a comparatively large, uncontrolled current through the unknown arm and relies on the arm being a passive resistor. A galvanometer is a delicate instrument whose coil would be overloaded and whose deflection mechanism is the thing being measured; its resistance must be found in a working low-current circuit — exactly what half-deflection does.
